1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In non-metastatic bladder urothelial carcinoma, TP53 status dictates the fate of everolimus-induced autophagy and thus determines therapeutic response, with the immune microenvironment acting as a context-dependent modifier. Specifically, TP53 wild-type tumors will convert mTORC1 inhibition-induced autophagy into cytotoxic autophagy, leading to reduced tumor viability and higher therapy response, especially when TIL density is high (notably CD8+ T cells). TP53-mutant tumors will exhibit either non-lethal autophagy or autophagy that fails to trigger cell death, resulting in attenuated or resistant therapy responses even in the presence of modest TILs. In TP53-wild-type tumors, high TIL density will correlate with improved everolimus response due to synergistic effects of autophagy-induced antigen presentation and immune-mediated killing; LAG3-expressing TILs will further modulate this effect by shaping T cell exhaustion and effector function, potentially blunting or delaying response unless LAG3-related pathways are targeted. In TP53-mutant tumors, the association between TIL density and response will be weaker, and LAG3 status will have a diminished impact on response unless combined with strategies that overcome autophagy-associated resistance. Primary endpoint: therapy_response (radiographic or pathologic) after everolimus exposure. Secondary endpoints: autophagy flux (LC3-II turnover, p62 degradation with lysosomal inhibition), mTOR pathway activity (p-S6K, p-S6), apoptosis markers (cleaved PARP/caspase-3), TP53 status validation, TIL density (CD3, CD8), and LAG3 expression on TILs. This hypothesis yields testable predictions across a two-factor framework (TP53 status: WT vs MUT; immune context: low vs high TIL density with LAG3 status) and supports exploration of combination strategies in TP53-MUT contexts (e.g., LAG3 blockade) to rescue therapy response.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	TP53, mTOR
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

TP53: TP53 with urothelial carcinoma (potential determinant of cell■fate decisions under stress).

mTOR: mTOR inhibition) induces autophagy that can promote cancer cell death in non■metastatic bladder (urothelial) carcinoma, yet clinical responses are heterogeneous — why do some tumors undergo autophagy-mediated death while others resist? 2) Tier 1 paths used: the mTOR → everolimus → autophagy → cell death axis (mechanism of drug action) and the association of TP53 with urothelial carcinoma (potential determinant of cell■fate decisions under stress).

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	LAG3	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: , can shape response to therapies that induce cellular stress and antigen presentation, making TP53- and immune-context a plausible axis for differential responses in non-metastatic BLCA. Studying this in non-metastatic disease reduces confounding systemic effects and supports biomarker-driven stratification.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- LAG3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.